

ForenSiX: Automated Network Forensics and Diagnostics for Beyond-5G and 6G Networks Using Large Language Models

Xuanbo Huang , Kaiping Xue , Lutong Chen , Jiangping Han , Jian Li , and David S. L. Wei 

ABSTRACT

Network forensics and diagnostics play a critical role in enabling operators to locate the root causes of attacks or service disruptions by analyzing logs and evidence of various devices. For example, when a subscriber reports a “disconnection” error, the network operator must investigate the subscriber’s service functionality and determine the underlying cause. However, these tasks are becoming more difficult in the future beyond-5G (B5G) and 6G environments. The expanding number of network functions, services, interfaces, and contexts results in more complex data flows and longer network management chains, making traditional manual diagnosis infeasible to trace and locate root causes from large-scale semantic network logs. To address these challenges, we propose ForenSiX, an automated framework designed for B5G and 6G networks. ForenSiX utilizes static analysis of relevant standards, module source code, and config files to construct a comprehensive data-dependency graph, enabling causal relationship analysis across multiple network functions. Next, it integrates a dynamic log-tracking subsystem that harnesses both regular expressions and commercial off-the-shelf large language models (LLMs). A multithreaded component further enhances processing efficiency, and a fault-tolerance unit ensures accuracy and robustness. We evaluate ForenSiX in Open5GS and UERANSIM, demonstrating its effectiveness, reliability, and adaptability in realistic scenarios.

INTRODUCTION

The fifth-generation (5G) networks have been commercially available since 2019, leveraging both sub- 5 GHz and millimeter-wave frequencies with peak data rates as high as 20 Gbps. From an architectural standpoint, 5G networks significantly enhance flexibility and scalability through service-oriented architectures and cloud-native designs [1]. In addition, advanced features such as network slicing and edge computing allow 5G to address diverse service requirements effectively. With the growing need for high-capacity connectivity, Internet of Things (IoT) use cases, and intelligent services, research and

implementation of next-generation networks are intensifying. The coming sixth generation (6G) will further advance, employing terahertz (THz) frequencies, AI-assisted network management, and converged communication and sensing [2]. 6G will deliver ultra-high-capacity, low-latency, and high-reliability services, thus supporting superior applications including energy-efficient networking, intelligent automation, and three-dimensional (3D) communication.

As networks become increasingly complex and feature-rich, their reliability and security become more critical than ever. Network forensics and diagnostics play a key role in troubleshooting the root causes of dropped or misrouted data, debugging malfunctioning network functions, and defending against cyber threats [3], [4]. For instance, when a subscriber encounters service disruptions – such as an inability to connect to the Internet or failure to meet expected Quality of Service (QoS) levels – network operators must diagnose the root cause. However, pinpointing the issue is challenging, as disruptions may arise from various factors, including inaccurate location data, faulty base stations, misconfigured core network components, or even security threats like hijacking attacks.

In 5G networks, the 3rd Generation Partnership Project (3GPP) has taken steps to standardize forensic analysis through TS32.421-TS32.423, which provides guidelines for logging critical data across network entities and protocols [5]. These specifications define the core forensic concepts, requirements, data formats, management methods, and key requirements for structured logging. However, forensic analysis in 5G remains largely manual, requiring experts to interpret log data to determine root causes. As networks grow more complex, this manual approach is increasingly unsustainable. In B5G and 6G, the number of network functions and service-based interfaces has expanded significantly. For example, the number of network functions has increased by 300% from Release-15 to Release-17, further complicating forensic analysis. This rapid growth emphasizes the importance of automated tracing and diagnostic techniques to analyze network behavior and improve security and reliability.

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To address these challenges, researchers have explored two primary approaches. The first applies machine learning to discover causal relationships between key performance metrics and their associated modules or nodes [6], [7]. While effective for attribution analysis, this method operates at a coarse-grained level, thus finer-grained dependency at the network function and context levels becomes tricky to discover. The second approach constructs fine-grained data dependency graphs with static and dynamic analysis [4], [8]. Although this method provides greater detail, this approach cannot comprehensively reflect network behavior and restore all contextual information of interest. Moreover, both approaches are not capable in real-time observation of network behavior and reacting to context changes from semantic logs of heterogeneous network functions. Thus, the construction of an efficient, automated diagnostic and forensic structure is still pending for next-generation networks.

Recently, commercial off-the-shelf large language models (LLMs) have drawn immense attention due to efficient processing of textual and semantic information. Researchers have actively explored LLM applications across various network management tasks [9], [10], [11], such as automated network configuration [10], routing table monitoring [9], and interactive network search [11]. However, their use for network forensics has received limited attention to date. In this context, integrating LLMs into forensic analysis offers promising opportunities to automate the log tracking and analysis process. Nevertheless, several practical challenges remain. First, LLM inference speed may not match the high log generation rates of network components. Second, LLMs may output contextual information and numerical values incorrectly, which could lead to forensic analysis errors. Any practical solution must address these challenges while keeping compatibility with existing forensic instruments.

In this paper, we propose ForenSiX, a framework designed to enable efficient and automated network forensics and diagnostics in the complex environments of B5G and 6G networks. ForenSiX integrates LLMs for forensic log analysis while introducing two dedicated mechanisms to overcome the above challenges. To address inference speed limitations, we design a multi-threaded scheduling component that dynamically adapts LLM query processing to meet log analysis demands. To mitigate potential inaccuracies, we introduce a fault-tolerance module that enhances the robustness and reliability of LLM-generated outputs. ForenSiX operates in two stages. First, building on prior work in forensics graph modeling [4], we establish a data dependency graph (DDG) that abstracts network data into nodes and edges, capturing entities, contexts, and events along with their relationships. This graph-based representation offers a structured and transparent view of network interactions. Second, to support dynamic event and context tracing, we implement an online tracking system that combines regular expression-based parsing with LLM-assisted semantic interpretation and real-time monitoring. Finally, we evaluate ForenSiX using Open5GS and UERANSIM. Although 6G implementations and simulators are not yet available,

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many foundational technologies are expected to build upon the 5G framework. Accordingly, we conduct experiments in a 5G environment, which serves as a practical and representative setting to assess the system's effectiveness and future applicability to 6G scenarios.

To summarize, this paper contributes the following:

Comprehensive Static Analysis: We conduct an extensive study of beyond-5G and 6G networks by analyzing relevant standards, source code, and configuration files. This static analysis produces a data dependency graph that supports causal relationship analysis among diverse network functions.

Dynamic Log Tracking System: We design a real-time log tracking framework that leverages both regular expressions and commercial off-the-shelf LLMs. The system includes a multithreaded scheduling component and a fault-tolerance module to enhance processing efficiency and robustness.

Empirical Evaluation: We validate ForenSiX using Open5GS and UERANSIM, demonstrating its effectiveness and resilience in realistic scenarios.

The remainder of the paper is organized as follows: The section “[Background and Related Work](#)” provides background information and discusses related work. The section “[ForenSiX System](#)” details the ForenSiX system design, while the section “[Evaluations](#)” presents the evaluation results. Finally, the section “[Conclusion](#)” concludes the paper.

BACKGROUND AND RELATED WORK

BACKGROUND

Figure 1 illustrates an overview of B5G and 6G networks, emphasizing key architectural components. The networks are structured in three layers: Infrastructure, Network Service, and Application. The Infrastructure Layer comprises essential physical components, including radio access networks (RAN), core networks (CN), satellite systems, switches, and routers. It provides the foundational resources required for network operations. The Network Service Layer is built on a cloud-native, service-based architecture, where the layer orchestrates network functions dynamically to enable efficient service deployment and management. It plays a critical role in supporting diverse applications by ensuring flexibility and scalability. The application layer integrates AI-driven automated network management and introduces emerging use cases, such as immersive smart cities. Finally, the Applications in this layer make use of the infrastructural and network services beneath to deliver advanced, intelligent solutions.

Typically, data from a user equipment (UE) is forwarded onto the Internet by a number of important components and processes. The UE first establishes a connection with a nearby base station, known as the gNB, which manages the

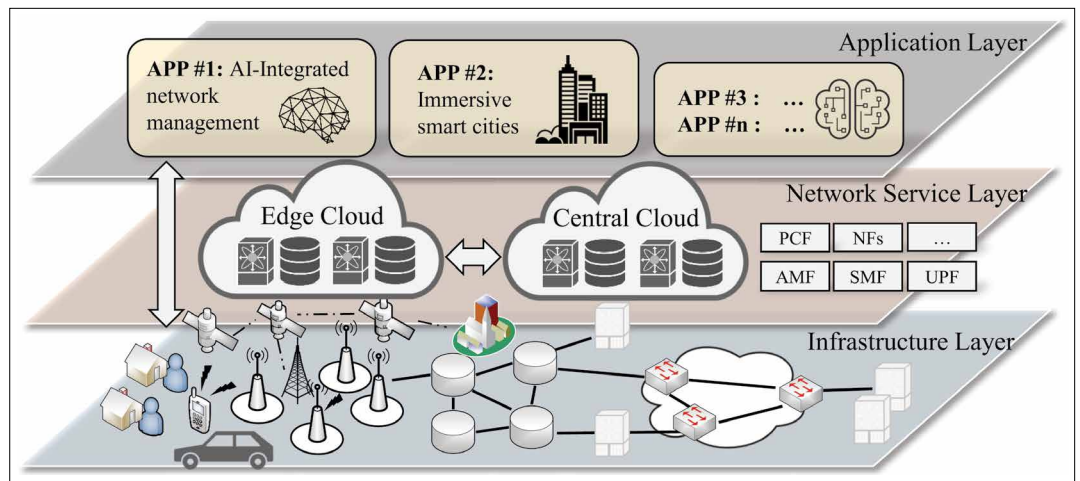


FIGURE 1. Overview of B5G and 6G network architecture, illustrating the Infrastructure Layer, Network Service Layer (with edge and central clouds), and Application Layer with representative services.

initial communication link. The gNB is responsible for handling radio resources and performing basic operations such as signal modulation and demodulation. Once the gNB receives data from the UE, it forwards it via a backhaul link to the core network (CN). Within the CN, the Access and Mobility Management Function (AMF) and the Session Management Function (SMF) handle connection management, session establishment, and mobility context. The AMF services the requests of authentication and access for users, the SMF deals with paths for data and allocates UEs' IP addresses, while the User Plane Function (UPF) also plays a vital role in leading and forwarding the radio access network (RAN) data packets into the corresponding data network, such as the Internet.

RELATED WORK

The security aspects of B5G networks have been extensively reported in the literature [12]. The integration of AI functions enhances attack detection and mitigation performance [13]. Advanced security frameworks, such as zero-trust architecture [14], [15], offer improved privacy and resilience. However, despite these improvements, network forensics and diagnosis in these evolving networks remain underexplored.

Network forensic tools are generally classified as host-level and network-level solutions [4]. Host-level tools monitor system logs to identify malicious activity at the operating system level and detect threats such as malware and viruses. Network-level forensic tools are further divided into packet-level forensics and flow-level forensics. While packet-level forensics reconstructs network events by analyzing data packets, it faces scalability challenges due to the massive number of users in B5G and 6G networks. In contrast, flow-level forensics has gained traction for its ability to process large-scale, complex network environments more efficiently.

Flow-level forensic analysis across different network planes has been an active research area, particularly with the adoption of software-defined networking [8]. Solutions such as ForenGuard [4] and PicoSDN [8] trace cross-plane events within SDN architectures to establish relational chains

for security incident detection and response. With B5G and 6G architectures, as they are increasingly heterogeneous and complicated, SDN-specific forensic solutions fall short. This is where the gap calls for an upcoming forensic solution that can handle the dynamic and complicated nature of B5G networks, giving rise to ForenSiX.

FORENSiX SYSTEM

In this paper, we introduce ForenSiX, a framework designed for forensic analysis and problem tracing in the complex landscape of B5G and 6G networks. ForenSiX facilitates network forensics by constructing a DDG that maps relationships among network entities, contexts, and events. The graph yields a visual representation of network activity and data flow. The system incorporates an activity tracker that gathers relevant data from a number of entities based on dependencies defined in the DDG. Finally, the diagnostic module provides the propagation chain of issue-related contexts and leaf nodes indicate potential root causes. Such a structured approach enhances troubleshooting efficiency and enables network operators to diagnose and resolve network issues.

MODELING

Our data model framework abstracts the network into three key components: Entity, Event, and Context. Entities include UE, gNB, and network function modules, such as AMF, SMF, UPF, and Artificial Intelligence-based Network Functions (AI-NF). ForenSiX requires each entity to have predefined, explicit network functions, which are typically specified by standardized implementation guides (e.g., for core network modules) or provided by source code (e.g., for AI-NF with new functionalities). Contexts are network states of the networks that evolve with time-based on events. A few examples include the NR state of UEs, PCF policy database, and UPF forwarding routing tables. These contexts can be modified by different network protocols and events, for instance, handover procedures, influencing the contexts of UEs, gNBs, and AMFs. Also, contexts influence the decisions of key network functions. For instance, load information from gNBs and UPFs notifies AI-NF about load balancing, which

consequently triggers respective context modifications. These kinds of event-context interactions are essential for network operations.

In order to enable ForenSiX functionality, we initially develop a data model dedicated to network forensics in 5G and 6G networks. The DDG is a directed graph $G = (V, E)$, with the node set V representing network entities, events, and contexts, and the edge set E representing their relationships (e.g., produce, read, write). For example, consider the context of a UE: a node in V represents this context. Nodes that are connected to this central node via incoming edges are entities that read the context—such as the UE, gNB, AMF, PCF, and potentially AI-based access network management modules. Similarly, nodes that have outgoing edges to the central context node are events that may trigger updates, such as UE registration, connection, or handover. Moreover, the entities causing such events are linked to the corresponding event nodes – for instance, a gNB causing a UE handover event.

SYSTEM DESIGN

As a network function module, ForenSiX is designed to perform three primary functions: model building, log collection, and fault analysis and localization for the relevant network entities. To achieve this, ForenSiX has three sub-modules: 1) DDG Generator: This module uses static analysis to analyze standard documents, extract entity, event, and context attributes, and construct the data dependency graph; 2) Activity Tracker: It captures logs from network modules to monitor context changes and observe events. The log analyzer then aggregates and stores these logs for further analysis; 3) Diagnosis Module: This module provides network operators with the capability to query logs and associated events through API interfaces, enabling forensic analysis and debugging. We will detail each module in the following sections.

DATA DEPENDENCY GENERATOR

In network forensics, it is essential to trace and analyze causal relationships across diverse log sources to identify the root causes of service disruptions. For example, when a user-reported issue occurs, the underlying malfunction may originate from multiple network entities or misconfigurations. To address this, the design objective of the DDG generator is to provide a comprehensive and structured view of causal relationships among network entities, contexts, and events, thereby supporting automated forensic analysis.

DDG construction is a mature technique widely adopted in the forensics domain [4], [8]. The workflow of the DDG generator consists of two main procedures: node construction and edge construction. After collecting input files, such as network function configuration files (entities), source code (contexts), and protocol standards (events). The DDG generation process includes the following steps:

- **Entity Node Construction:** For each entity, we apply static analysis [4] to extract basic information such as entity names, IP addresses, and API endpoints to establish entity nodes. For example, in Open5GS,

this information is stored in `YAML` format files. ForenSiX can also invoke the Network Function Repository Function (NRF) via its RESTful API to retrieve the registered NFs and their IP addresses in a running network.

- **Context Node Construction:** Network function contexts can be categorized into two types. The first type consists of standardized contexts, such as SUPI and GUTI for UEs, which have clear definitions in standards and can be extracted using automated tools [4], [8]. The second type consists of implementation-specific contexts embedded in network modules. ForenSiX extracts structured data from 3GPP PDF documents using `pdfplumber`, and parses header files (.h) with regular expressions to extract declarations and struct names. We further link the context nodes to entity nodes using corresponding module names.
- **Event Node Construction:** We extract structured event lists from standard documents, such as TS 32.423 [5], using `pdfplumber`, and establish event nodes accordingly. For example, TS 32.423 provides signaling event lists and related context names in structured tables. These events are extracted as nodes and connected to the corresponding context nodes to form edges.

After completing node and edge construction, the DDG is finalized. Figure 2 visually distinguishes these elements using different shapes: rectangles for contexts, diamonds for events, and circles for entities. In the directed graph, an edge from node A to node B indicates a context propagation path. For instance, a UE's context (e.g., TAI/Cell ID) is linked to a "mobility update" event, which subsequently connects to the AMF's UE list context. This indicates that the mobility update event reads the TAI/Cell ID and updates the UE's mobility information stored in the AMF.

ACTIVITY TRACKER

After constructing the DDG, the activity tracker must monitor network changes in various logs and construct a tracking tree structure for later diagnosis. This task is challenging because semantic logs cannot be easily extracted or directly mapped to DDG elements. To address this, we

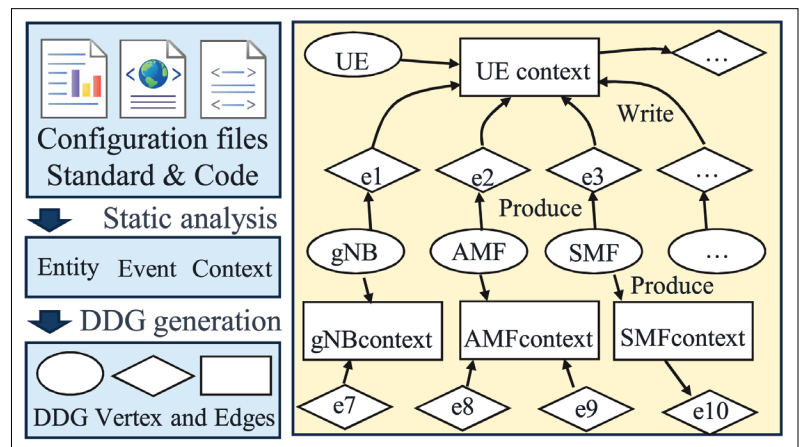


FIGURE 2. Overview of the DDG generator: The data dependency graph captures the relationships among network entities, contexts, and events.

To process timely and keep pace with network log generation speed, ForenSiX constructs a new thread to process the LLM query when the buffer reaches the optimal number per query, allowing the system to adjust speed and effectiveness dynamically.

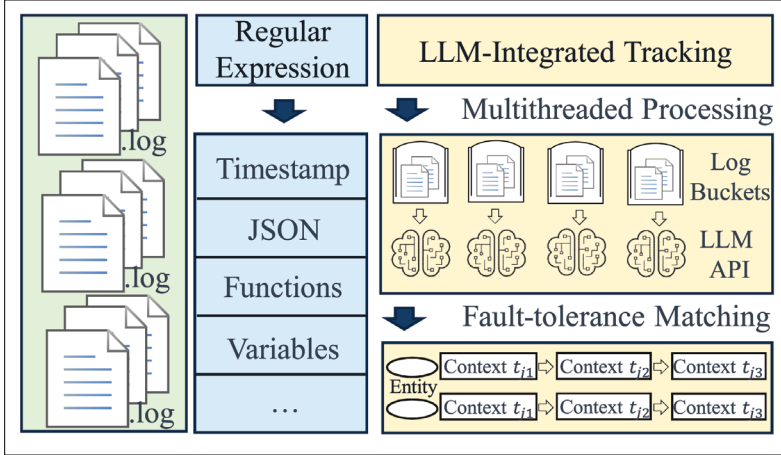


FIGURE 3. Overview of the activity tracker. The activity tracker collects logs from multiple entities and processes them using regular expressions and LLMs.

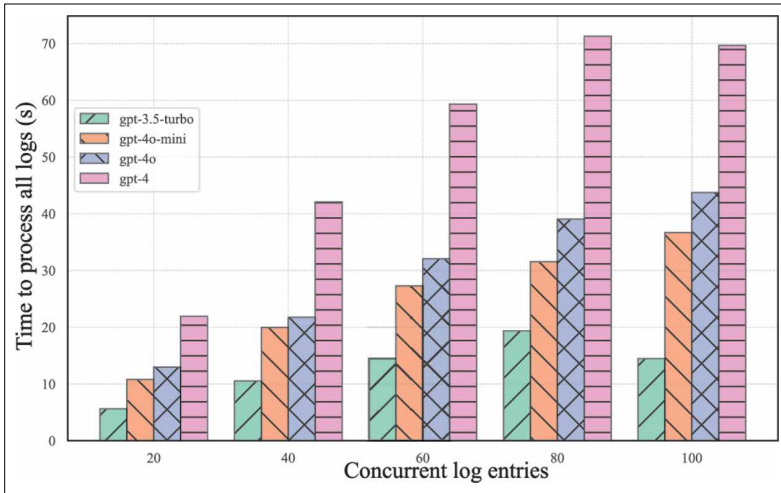


FIGURE 4. Processing time of LLMs for varying log entries number per request.

design a tracking system that combines regular expressions with commercial off-the-shelf LLMs to monitor events and contexts in real-time. However, integrating LLMs into log analysis presents two practical challenges.

Because data is processed more slowly by LLMs than it is being generated in the logs, we develop a multithreaded component to mitigate this limitation. Then, LLMs can lead to errors in understanding the logs. To reduce errors, we introduce a fault-tolerance module, which makes log analysis more dependable.

ForenSiX pulls out the log files of each network function through trace sessions [5]. Figure 3 illustrates the activity tracker workflow. First, well-structured log components — such as timestamps, function names, log locations, types, and states written in Screaming Snake Case — are extracted using regular expressions. ForenSiX can also support other formats of well-structured

components without requiring major changes. Second, for context-related logs (e.g., logs from context . . . c) that cannot be processed using regular expressions, ForenSiX delegates the task to the LLM, requesting it to extract the corresponding context and value changes from semantic sentences. To achieve this, ForenSiX prebuffers log records until a threshold is reached before sending them to the LLM. The threshold value is discussed later to balance performance and accuracy.

If the number of logs in a query is excessively high, the LLM will struggle with long-text processing and incur more errors. Conversely, sending logs individually incurs network latency and processing overhead. To process timely and keep pace with network log generation speed, ForenSiX constructs a new thread to process the LLM query when the buffer reaches the optimal number per query, allowing the system to adjust speed and effectiveness dynamically. Moreover, even in the case where LLM's temperature is specified as 0 to ensure deterministic output, the LLM answers still show diversity due to small differences in tokenization, hardware execution, and tie-breaking policy. To counter this, ForenSiX employs a fault-tolerant approach with fuzzy matching of LLM output and DDG elements. After parsing the logs, ForenSiX extracts the response and parses the context. It does not use an exact match but computes the Levenshtein distance between the parsed context and the DDG elements and selects the most similar match as the new element.

DIAGNOSTIC MODULE

After the activity tracker constructs the complete event traceback tree, this structure is presented to the network operator for further analysis. The relevant context may include multiple distinct traceback chains, each node representing an event that affected the context and the entity that initiated the event (the forward propagation path). Meanwhile, nodes to which the context points represent the backward propagation path. For instance, if an UE reports difficulty connecting to the data network, the activity tracker displays all event chains influencing the context around the reported time.

EVALUATIONS

We evaluate our system's performance using Open5GS and UERANSIM. Because 6G network implementations and simulators are not yet available, and many fundamental 6G technologies are expected to build upon the 5G framework, using a 5G environment provides valuable insights into 6G networks. Therefore, we rely on 5G-based evaluations as a practical alternative. All experiments are conducted on an Ubuntu 22.04 machine equipped with 200 GiB of memory and a 98-core Intel Xeon CPU. We deploy Open5GS as the 5G core network and use UERANSIM to simulate ten users connecting to the core network and transmitting data randomly. Each experiment collects ten minutes of log data from the 5G core network, with each data point repeated 100 times to obtain an average.

First, we evaluate the processing time for different numbers of log entries using commercial off-the-shelf LLMs. In this experiment, our system calls the OpenAI API to process the logs

and records the response time for each request. Figure 4 presents the total processing time for different models when handling various numbers of log entries (ranging from 20 to 100). The average processing time per log entry is shown in Figure 5. As expected, the processing time increases as the number of log entries grows. Notably, the GPT-4 model reaches a bottleneck at around 100 log entries. Additionally, the increase in log entries leads to a decrease in the average processing time per entry, which is a critical metric for real-time log analysis systems. For the advanced GPT-4o and GPT-4o-mini models, when processing up to 100 log entries, the average processing times per log entry were 0.437 and 0.366 seconds, respectively. In contrast, GPT-4 took more than 0.7 seconds. It is also worth noting that while the GPT-3.5-turbo model had the fastest processing time (0.145 seconds per log entry), its low accuracy (discussed later) makes it unsuitable for production environments.

We evaluate the accuracy of LLMs when processing different numbers of logs. In this experiment, we define a log as accurate if the LLM output meets two conditions: 1) The LLM must correctly extract the context name (either exact or tolerant match); 2) The LLM must correctly extract the context value (exact match). Figure 6 compares the effectiveness of our tolerant matching method against the raw output from the LLM. As discussed earlier, due to subtle token variations in the LLM input, the model may produce multiple semantically equivalent results when identifying contexts. This necessitates the use of the tolerant match to merge such results. As shown in Figure 6, our method consistently improves accuracy by approximately 15% to 20% across multiple models. In the case of GPT-4o, it ensures an accuracy above 98% for any input length. Thus, our tolerant matching method enhances the feasibility of using LLMs for log analysis and is universally applicable across various models.

DISCUSSION

ForenSiX does not address all challenges in this domain. In the following, we discuss several important research directions that remain open for future exploration. In particular, we identify three important directions that warrant continued investigation: the development of forensics-specialized LLMs, enabling intent-aware analysis, and addressing data privacy and security concerns.

First, one important direction is developing forensics-specialized LLMs that combine general linguistic competence with comprehensive protocol-level understanding. It could be achieved through fine-tuning models on tagged network events or using formal protocol definitions as part of the training data. An LLM attuned to network vocabulary would be better suited to automate forensic examination at scale, from low-level logs to making sense out of them.

Second, intent-aware analysis represents an opportunity. Traditional forensic tools output raw facts and are unable to infer context or intent. LLMs can be semantic engines that infer the narratives behind observed events. Future forensic pipelines could employ LLMs to connect structured outputs of intrusion detection systems or log collectors to create high-level conclusions. For

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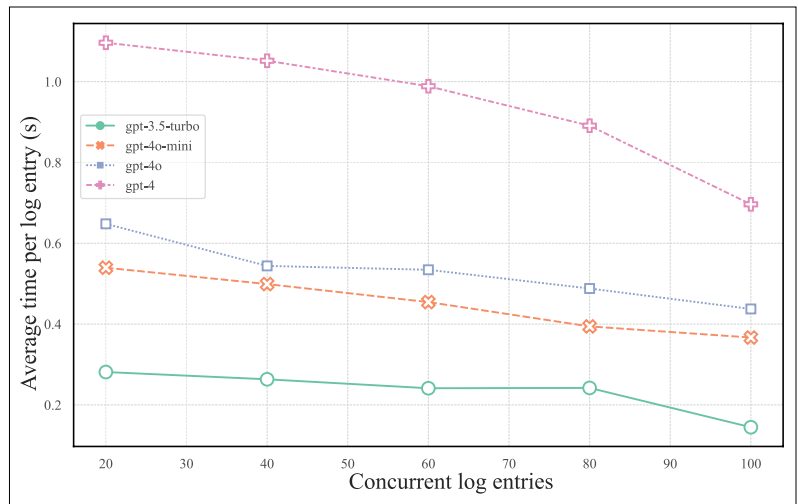


FIGURE 5. Average processing time per log entry for LLMs at varying numbers of log entries per request.

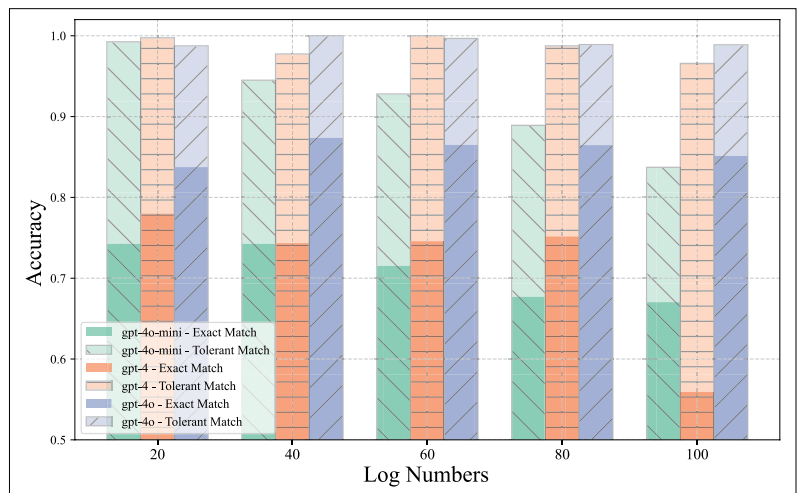


FIGURE 6. Accuracy of the tolerant matching mechanism for different LLMs at varying numbers of log entries per request.

example, the recognition of anomalous login followed by file deletion as indicative of an insider threat. By reasoning over entities, contexts, and event orders, LLMs can help advance forensics beyond shallow pattern matching to richer narrative-style analysis.

Third, data privacy and security concerns are always paramount when using LLMs in forensic workflows. Network forensic data are often sensitive information, and sending them to commercial off-the-shelf LLM services might potentially cause compliance and confidentiality concerns. Current research has also shown that large models possess the unintended ability to memorize training data, potentially leaking sensitive information in future output. To mitigate these risks, upcoming work is expected to explore local deployment approaches, including on-premises or fully sovereign open-source LLMs. While these precautions

help safeguard sensitive information, they also introduce new issues, including imposing sufficient access controls, auditing AI-generative outputs, and maintaining model security.

CONCLUSION

In this paper, we propose ForenSiX, a semi-automated diagnosis framework for 5G and 6G networks. ForenSiX supports network operators in diagnosing the root causes of network failures through a user-friendly data dependency graph. The graph traces network context propagation between events and entities and therefore enables efficient and accurate diagnosis. Furthermore, ForenSiX uses commercial off-the-shelf LLMs to analyze the semantic logs of diverse network components in real-time. It achieves this with multithreaded query components and a fault-tolerance matching module. We evaluate ForenSiX in Open5GS and UERANSIM, demonstrating its effectiveness in realistic scenarios.

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